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Q1. Program based on Classes and Objects

Score: 5.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;

class Main{
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        Student s = new Student();

        s.rollNo = sc.nextInt();
        sc.nextLine();
        s.name = sc.nextLine();
        s.marks = sc.nextFloat();

        s.displayDetails();
        sc.close();
    }
}

class Student {
    int rollNo;
    String name;
    float marks;

    void displayDetails(){
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }
}
```

INPUT

```
12
Sai
88.5000
```

OUTPUT

```
Roll No: 12
Name: Sai
Marks: 88.5
```

Q2. Initialize Object Data Using Setter Methods

Score: 5.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Student {
    private int rollNo;
    private String name;
    private double marks;
    public void setRollNo(int rollNo)
    {
        this.rollNo = rollNo;
    }
    public void setName(String name){
        this.name = name;
    }
    public void setMarks(double marks){
        this.marks = marks;
    }
    public void displayDetails(){
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }
    public static void main(String[] args){
        Scanner sc = new Scanner(System.in);
        Student s = new Student();
        s.setRollNo(sc.nextInt());
        sc.nextLine();
        s.setName(sc.nextLine());
        s.setMarks(sc.nextDouble());
        s.displayDetails();
        sc.close();
    }
}
```

INPUT

```
1120
Rajesh
87.5
```

OUTPUT

```
Roll No: 1120
Name: Rajesh
Marks: 87.5
```

Q3. Student Details Using Default and Parameterized Constructors

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Student {
    int rollNo;
    String name;
    double marks;
    Student(){
        rollNo = 101;
        name = "Rahul";
        marks = 85.5;
    }
    Student (int rollNo, String name, double marks){
        this.rollNo = rollNo;
        this.name = name;
        this.marks = marks;
        }
    void displayDetails(){
        System.out.println("Roll No: " + rollNo);
        System.out.println("Name: " + name);
        System.out.println("Marks: " + marks);
    }
    public static void main(String[] args)
    {
        Scanner sc = new Scanner(System.in);
        Student s1 = new Student();
        s1.displayDetails();
        int rollNo = sc.nextInt();
        sc.nextLine();
        String name = sc.nextLine();
        double marks = sc.nextDouble();
        Student s2 = new Student(rollNo, name, marks);
        s2.displayDetails();
    }
}
```

INPUT

```
77
chandana
97.6
```

OUTPUT

```
Roll No: 101
Name: Rahul
Marks: 85.5
Roll No: 77
Name: chandana
Marks: 97.6
```

Q4. Net Salary Calculation Using Single Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Employee {
    int empId;
    String empName;
    double basicSalary;
    void getEmployeeDetails(int empId,String empName,double basicSalary){
        this.empId = empId;
        this.empName = empName;
        this.basicSalary = basicSalary;
    }
public static void main(String[] args) {
    Scanner sc = new Scanner(System.in);
    int empId = sc.nextInt();
    sc.nextLine();
    String empName = sc.nextLine();
    double basicSalary = sc.nextDouble();
    Salary s = new Salary();
    s.getEmployeeDetails(empId,empName,basicSalary);
    s.calculateSalary();
    s.displayDetails();
    sc.close();

}
}
class Salary extends Employee{
    double hra,da,netSalary;
        void calculateSalary(){
           hra = basicSalary * 0.20;
           da = basicSalary * 0.10;
           netSalary = basicSalary + hra + da;
       }
       void displayDetails(){
       System.out.println("Employee ID: " + empId);
       System.out.println("Employee Name: " + empName);
       System.out.println("Basic Salary: " + basicSalary);
       System.out.println("HRA: " + hra);
       System.out.println("DA: " + da);
       System.out.println("Net Salary: " + netSalary);
    }
}
```

INPUT

```
1023
Ravi
50000
```

OUTPUT

```
Employee ID: 1023
Employee Name: Ravi
Basic Salary: 50000.0
HRA: 10000.0
DA: 5000.0
Net Salary: 65000.0
```

Q5. Employee Gross Salary Calculation Using Multilevel Inheritance

Score: 10.00 **TC: 3/3 passed**

CODE

```
import java.util.Scanner;
public class Employee{
    int empId;
    String empName;
    void getEmployeeDetails(int empId,String empName){
        this.empId = empId;
        this.empName = empName;
     }
public static void main(String[] args){
    Scanner sc = new Scanner(System.in);
    int empId = sc.nextInt();
    sc.nextLine();
    String empName = sc.nextLine();
    double basicSalary = sc.nextInt();
    GrossSalary g = new GrossSalary();
    g.getEmployeeDetails(empId,empName);
    g.getBasicSalary(basicSalary);
    g.calculateSalary();
    g.displayDetails();
    sc.close();
   }
}
class Salary extends Employee {
     double basicSalary;
     void getBasicSalary(double basicSalary){
         this.basicSalary=basicSalary;
     }
}
class GrossSalary extends Salary{
  double hra,da,grossSalary;
  void calculateSalary(){
     hra = basicSalary * 0.20;
     da = basicSalary * 0.10;
     grossSalary = basicSalary + hra + da;
     }
  void displayDetails(){
      System.out.println("Employee ID: " + empId);
      System.out.println("Employee Name: " + empName);
      System.out.println("Basic Salary: " + basicSalary);
      System.out.println("HRA: " + hra);
      System.out.println("DA: " + da);
      System.out.println("Gross Salary: " + grossSalary);
  }
}
```

INPUT

```
102
Chand
60000
```

OUTPUT

```
Employee ID: 102
Employee Name: Chand
Basic Salary: 60000.0
HRA: 12000.0
DA: 6000.0
Gross Salary: 78000.0
```